PRIVOKE: A Client-Side Semantic Privacy Firewall for User Prompts in Web-Hosted Large Language Models

Nicholas Bisset, Manpreet Sahadevan, Nixon Ngai, and Sanoop Mallissery

***Abstract*—Useful prompts often contain context that a user may not intend to disclose to an external language-model provider. PRIVOKE places local inspection and explicit allow, warn or block actions before transmission on supported client paths. Its lifecycle connects request enforcement to a separate training service, which checks version and retention constraints before distributing accepted parameters; independently randomized telemetry supplies no training data. On 1,560 authored provisional cases, updated packages improved contextual policy agreement, while targeted detector removal exposed a trade-off between unnecessary restrictions and missed protection. A frozen 498-row semantic annotation-presence check found specificity of only 13.36 to 21.76%. Thirty-three installed receiver probes verified declared request behavior. Every matched full-model candidate improved aggregate agreement yet failed retention. Installed pretrained-model measurements isolated resident detector-activation cost, while a quality-tier probe blocked a benign prompt. Controlled telemetry operation verified delivery and accounting; privacy audits found a small excess over nominal loss and substantial synthetic estimation error. Real-user monitoring usefulness remains unmeasured. These findings support an inspectable protection and revision mechanism while leaving independent contextual accuracy, useful adaptation without lost protection and practical user benefit unresolved.**

***Index Terms*—Local differential privacy, local inference, model regression, prompt privacy, randomized telemetry, request enforcement.**

I. INTRODUCTION

WEB-HOSTED large language models (LLMs) support writing, coding and analysis by processing the instructions and contextual material that users provide [1]–[3]. A useful prompt may therefore contain much more than a question: a user seeking help with a report might include a colleague's details, health information or confidential organizational material. The same context that helps the model complete the task can consequently disclose information to an external provider. Reported workplace incidents illustrate how this can occur during otherwise legitimate work [4], [5]. Controls over subsequent storage or use do not prevent the initial transfer. Inspecting the request on the user's device provides an opportunity to intervene while the information is still under local control.

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GitHub Repository Link.

Deciding when to intervene requires more than recognizing names or identifiers. Two prompts can discuss the same subject while differing in whose information they contain, who is intended to receive it and whether that disclosure is appropriate. Nissenbaum's account of contextual integrity similarly relates privacy to the norms governing information flows within a particular context [6]. A detector can represent some of these distinctions, but its labels do not independently establish authorization. Its decision must also have an observable effect on the request: notifying a user that content may be sensitive is different from preventing that content from leaving the device. These considerations motivate evaluating contextual predictions, policy decisions and actual transmission separately.

Motivated by this problem, we present PRIVOKE, an implemented client-side privacy firewall for supported outgoing prompts. The local detector combines rules, named entity recognition (NER) and semantic classification, describing information through sensitivity, intended audience and categories before selecting ALLOW, WARN or BLOCK. The extension releases the original request for ALLOW and WARN and cancels a supported request for BLOCK. WARN attempts to display a notice without waiting for acknowledgement, so it does not establish consent or prevented disclosure. This explicit request behavior allows the evaluation to distinguish a correct classification from an intervention that actually stops transmission.

Maintaining this protection introduces a second problem: a revised model can improve average accuracy while becoming wrong on cases that its predecessor handled correctly [7]. PriVoke therefore separates training a candidate from accepting it for use. A server-side runtime learns from supplied labelled examples, and the update service checks the candidate against its exact parent model and declared retention criteria before publication. Clients verify accepted artifacts and refresh their local parameters without an application release. A third, independent path randomizes coarse event reports on the workstation. Neither intercepted prompts nor telemetry supplies training labels, allowing detector maintenance and reporting to be examined without treating collected user traffic as supervision.

Local protection and learned sanitization already have substantial precedent in Casper, Hide and Seek and ProSan [8]–[11]. Our contribution is an implemented lifecycle and an evaluation that distinguishes three decisions: whether a request should leave the client, whether a revised detector should

replace its predecessor, and whether randomized reports support useful monitoring. Execution records connect classifications to request outcomes; version-bound comparisons expose candidate gains and harms; separate resource and telemetry measurements delimit the cost and information released. These are contributions in integration and empirical understanding, not claims to invent local detection, regression analysis or randomized response.

We organize the evaluation around three questions:

1) **RQ1: Does PriVoke detect and enforce privacy-sensitive prompt policies effectively?**
2) **RQ2: Can PriVoke update its detector without sacrificing previously protected behavior?**
3) **RQ3: What systems and privacy overhead does PriVoke introduce?**

RQ1 contributes a connected account of contextual prediction and request enforcement, including cases where better classification or fewer restrictions do not imply better protection. RQ2 contributes matched evidence about whether aggregate improvements survive a casewise retention requirement. RQ3 quantifies the measured resource boundary and separates report-content privacy from estimation and contribution-selection error. Together, these answers identify which conclusions the implemented lifecycle supports and which require evidence beyond successful local execution, training or accounting. The following sections develop that argument while retaining the limits of provisional contextual labels.

II. Related Work and Contribution

Local intervention and contextual decisions. Casper demonstrates how prompt protection can be placed on the user's device through a browser extension combining rules, entity recognition and local topic analysis [8]. Its reviewed prototype replaces identifiers with placeholders, supports user-triggered restoration in responses and requires acknowledgement of topic warnings. Analysis is initiated through a keyboard shortcut. Casper also permits custom rules and topics and compatible model replacement, establishing precedent for both layered local inspection and configurable protection. PriVoke builds on that motivation while examining a different request contract: its supported hooks automatically apply a decision, but a WARN releases unchanged content without waiting for the user.

Hide and Seek (HaS) addresses the usefulness of protected interactions by hiding entities before cloud inference and recovering them in the response [10]. Its generative variant trains hiding and recovery models, while the label-based variant uses an entity recognizer for hiding. ProSan instead uses task-importance and privacy proxies to select word replacements, then trains a sanitizer for local execution [11]. PromptShield considers a related enterprise setting through role/policy classification [12]. These works distinguish detection, disclosure and task completion. PriVoke's labels represent its policy, but do not independently validate contextual authorization.

Retaining behavior during model revision. The distinction between better average performance and preserved individual predictions is established in model-update research. Yan et al. study *negative flips*, where an updated image classifier becomes incorrect on a previously correct example, and introduce positive-congruent training to reduce such regressions [7]. Their result motivates measuring both gains and losses, although image-classification evidence does not establish privacy-policy correctness. PriVoke examines the related engineering decision of whether to publish a candidate: label and action outcomes are compared with the declared base, and a configured retention guard can reject an otherwise improved model. Binding that decision to a versioned artifact and verified client refresh makes acceptance distinct from successful optimization. The contribution is the evaluated connection, not new regression analysis or training; the guard covers only declared cases/criteria.

Privacy of operational reporting. Local randomization provides a foundation for collecting aggregate information without sending each underlying value directly to a collector. RAPPOR applies randomized response to client-side statistics and explicitly addresses the additional privacy cost of repeated reporting [13]. PriVoke uses categorical event reports with a local budget ledger, rather than adopting RAPPOR's memoized reporting mechanism. Its relevant question is therefore whether the implemented sampler and accounting satisfy their stated event-content bounds, and whether the resulting estimates are useful. We examine finite arithmetic, estimation error and contribution selection separately. The reporting mechanism does not protect participation or network metadata, and aggregate event fields supply neither corrected contextual labels nor evidence that the detector should be retrained.

Contribution and comparison boundaries. Sanitization and recovery assess disclosed content and continued use; negative-flip analysis concerns changed predictions; randomization constrains released observations. These outcomes require different acceptance evidence. RQ1 connects a layered decision to receiver-observed transmission without claiming Casper's entity/topic, HaS's recovery-resistance or ProSan's task-utility results. RQ2 examines whether aggregate gains satisfy version-bound retention, separating optimization from permission to publish. RQ3 separates finite content privacy, estimator error and admission selection alongside scoped cost. The implemented lifecycle makes these decisions jointly inspectable and measures where they diverge. The primitives are established; incompatible denominators preclude numerical superiority.

III. System Design and Privacy Boundary

A. Scope and Trust

PriVoke is designed to intervene when personal, third-party or organizational information is about to leave the workstation through a supported prompt request. The enabled extension, local runtime/supervisor, workstation and authenticated publisher form its trusted boundary. Compromise or deliberate replacement of page hooks can defeat protection; blocking a later request cannot recover information already disclosed.

The implemented interception path recognizes prompt POST formats submitted through page `fetch` and asynchronous `XMLHttpRequest` (XHR). Workers, attachments,

other transports and changed endpoints require their own validation; support for the two hooks does not establish coverage of those paths. Users can also disable protection, which bypasses inspection by design. An optional hosted-inference configuration sends text beyond the workstation and therefore has a different disclosure boundary. It is excluded from the local-inference measurements in this paper.

Fig. 1 separates prompt inspection, supplied training examples and randomized reporting. Only a permitting action releases the prompt to the hosted LLM. Detector maintenance therefore need not collect user prompts, and the collector supplies no training labels. Appropriate decisions remain an empirical question.

B. Local Inspection and Request Enforcement

The page hook extracts a supported prompt and sends it through the extension background and loopback gRPC-Web remote-procedure-call bridge to the workstation Python runtime. Native messaging starts the managing supervisor. The returned action releases or cancels the original request; the separate training runtime is not an inference fallback.

The runtime first normalizes the text while retaining offsets that connect its findings to the original input. It then evaluates regular-expression rules before the enabled NER and semantic layers. A rule-based BLOCK ends the cascade immediately. Otherwise, the findings are combined using the strongest action. This ordering has a direct consequence for interpretation: a semantic classifier cannot reverse an earlier rule-based restriction, and an early BLOCK means that the later requested detectors did not run.

Failure handling follows the same need to identify what was actually assessed. A failed requested layer changes an otherwise ALLOW result to BLOCK, while valid non-ALLOW findings remain visible. If analysis itself is unavailable, the extension blocks supported requests. Losing the parameter stream can instead leave rules and NER available as a reduced detector configuration. That fallback may release a request despite losing semantic coverage, so a returned action must be interpreted together with the recorded layer execution.

The semantic classifier represents a prompt through sensitivity, audience and information categories. Sensitivity distinguishes no identified risk (S0), low-risk personal context (S1), potentially harmful personal information (S2), and highly sensitive information, identifiers or secrets (S3). Audience, called *visibility* in the implementation, comprises public (P0), semi-public (P1), authentication-restricted (P2), group-private (P3), personal-private (P4) and unspecified (PU) contexts. The ten categories are health, politics, religion, criminal, financial, sexual, child, location, identity and third-party information; several may apply together.

The policy normally assigns BLOCK to S3 and WARN to S2. S0/S1 normally allow, except that identity or location triggers WARN when both categories occur together or visibility is P2–P4. S0 with no categories always allows. A numeric confidence below 0.5 overrides this mapping: S3 becomes WARN, S2 remains WARN only with identity or location, and all other cases become ALLOW. Confidence summarizes sensitivity/category heads and is not a calibrated probability of joint correctness or authorized disclosure. Neither the labels nor this deterministic mapping has been independently validated as a model of contextual authorization.

For illustration, assume S2, HEALTH and confidence 0.7, with no earlier restriction. The policy selects WARN and the supported request sends its original body without awaiting acknowledgement. An earlier rule BLOCK instead cancels the request before semantics runs. This hypothetical traversal applies the mapping; it is not an observed prediction, independently labelled case or additional test.

The models performing this task are classifiers, not conversational response generators. Historical V1 classifiers have randomly initialized frozen encoders and fitted heads; V3 packages retain pretrained Bidirectional Encoder Representations from Transformers (BERT) encoders and poolers [14]–[17]. Both offer efficient, balanced and quality packages, without a guaranteed performance ordering. The standard installed default is now selected V3 balanced, with explicit V3 tiers and labelled older V1 choices. Its native CPU loader verifies binary packages from the model service without a random or V1 fallback. Tokenizers, capacities, thresholds and training histories differ as well, so the comparison measures the resulting package change rather than the isolated effect of pretraining or architecture.

C. Guarded Detector Updates

The update path asks two separate questions: whether a candidate belongs to the intended model, and whether its behavior meets the configured acceptance criteria. A controller prepares bounded labelled batches and held-out guard cases, excluding protected rows, texts and groups from training. The server runtime loads the specified base, trains a candidate and returns parameter changes and metrics tied to that base. Before publication, the updater checks base identity, permitted tensors and shapes, finite values, magnitude limits and the strategy's configured guards. Artifact checks establish that the candidate is well formed; retention checks assess its behavior on declared cases. Completing a fit or transferring its parameters cannot substitute for either decision.

An accepted candidate is published through an atomic compare-and-swap operation, which requires the current base still to match its declared parent. This prevents a decision made for one model from silently replacing a different intervening revision. Durable replay receipts associate an identical retry with its earlier artifact outcome instead of applying it twice. Head and underlying-model stages are separate publications, however, so rejecting a later stage does not undo an accepted head update. Clients verify delivered artifacts before refreshing cached weights. Standard V3 uses a pinned read-only package store and configuration-owned alias; the separately enabled legacy training/publication route cannot replace it. Unsupported V3 legacy-gradient calls are rejected. Installed V3 inference therefore does not establish deployed V3 learning or guarded publication.

Centralized training uses supplied examples, not user gradients or federated learning. An implemented service path is not evidence of cloud deployment or a revision worth retaining.

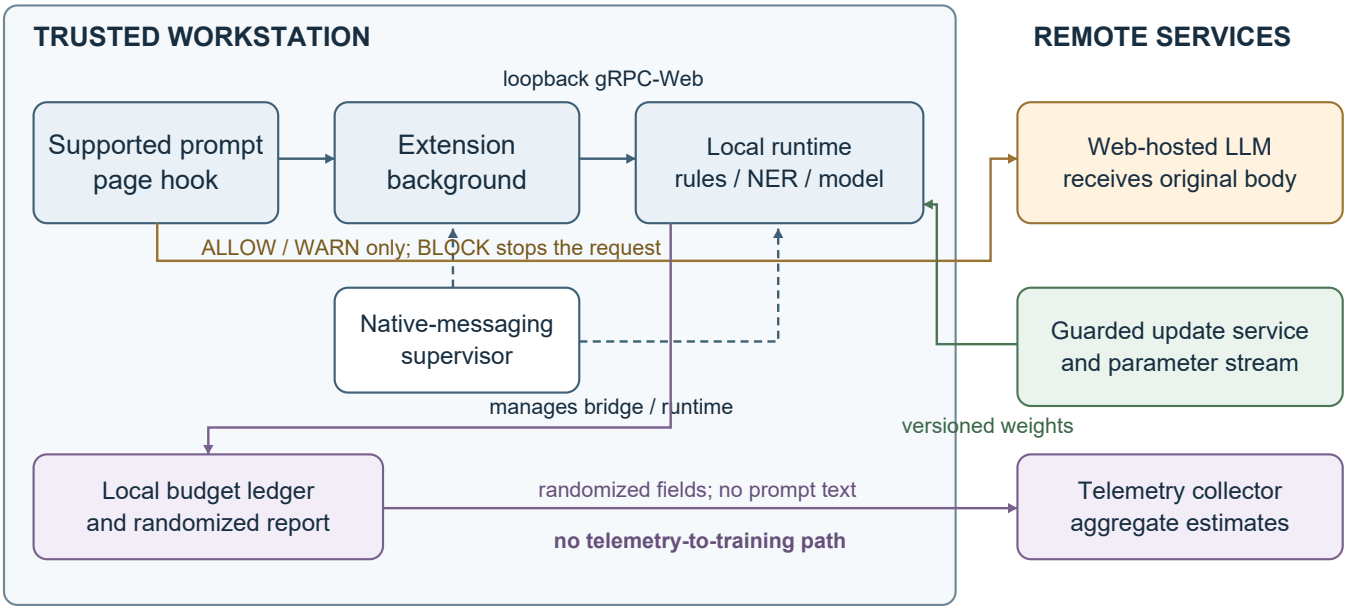


Fig. 1. PriVoke's three data paths. Supported prompt text is inspected inside the trusted workstation. Only an ALLOW or WARN decision releases the original body to the web-hosted LLM. Native messaging manages the local bridge/runtime; analysis uses loopback gRPC-Web. A separate update service supplies versioned weights, and a separate collector receives randomized event fields. Training consumes supplied labelled examples, not intercepted user prompts or telemetry. Dashed lines denote lifecycle control.

D. Separate Randomized Telemetry

Telemetry describes coarse aspects of an event through five fields: action, an action/sensitivity risk bucket, one category selected by fixed precedence, a bounded release label and a four-hour time-of-day bin without a date. Their domain sizes are respectively $K_i = 3, 4, 11, 2, 6$; the category domain includes the no-category value. The workstation randomizes each field before release. Reports omit raw prompts, identifiers, exact confidence scores and per-layer findings, reducing the information presented to the collector without making the retained fields intrinsically harmless. The collector estimates lifetime marginals for usage/development inspection, not latency or error rates. It has no analysis/user denominator, date/window query or joint-event record; its version domain is only v0.3.0/OTHER. A separate Health call checks store writability. Neither channel supplies corrected training labels.

The intended local differential privacy guarantee limits how much the probability of an output can change between possible event contents [18]. Here the comparison holds the reporting opportunity and public configuration fixed; it does not hide whether an event was reported. An event budget ε is divided equally, giving each field $\varepsilon_i = \varepsilon/5$. A field with K_i possible values reports its true value with probability p_i and each alternative with probability q_i:

$$p_i = \frac{e^{\varepsilon_i}}{e^{\varepsilon_i} + K_i - 1}, \quad q_i = \frac{1}{e^{\varepsilon_i} + K_i - 1}. \tag{1}$$

With independent ideal samples, multiplying the field likelihood ratios gives an event bound of e^{ε}, following composition [18]. The implemented sampler instead compares a fresh uniform 53-bit integer draw, $J/2^{53}$, with a binary64 threshold $\widetilde{p}_i$ and uses rejection sampling to choose an alternative. Its finite truth and alternative probabilities and event loss are

$$P_i = \frac{\lceil 2^{53}\widetilde{p}_i \rceil}{2^{53}}, \quad Q_i = \frac{1 - P_i}{K_i - 1}, \qquad \varepsilon_{\text{finite}} = \sum_i \log \max\{P_i/Q_i, Q_i/P_i\}. \tag{2}$$

This calculation conditions on the host arithmetic and independent uniform random bits; the ideal formula alone does not certify the implementation. A persistent ledger reserves budget before construction, with defaults of one per event and eight per Coordinated Universal Time (UTC) day. Construction or delivery failure does not refund a reservation. Retained state and a trusted clock are necessary for accounting. Report presence, contribution selection, delivery and network metadata remain outside the content guarantee. To estimate marginals, the collector inverts nominal response probabilities and clips counts to their feasible range. Whether those estimates remain useful after randomization and clipping is a separate empirical question.

IV. Evaluation Method

A. Questions and Evidence Units

Table I connects prediction and enforcement, retained behavior, and systems/reporting costs to the three RQs. Several analyses reuse cases, so their agreement is not independent confirmation.

Product calls request rules, NER and semantics [2,3,4], recording legitimate skips after rule BLOCK. Sole-semantic

TABLE I
RETAINED EVIDENCE AND ITS INFERENTIAL SCOPE. CONTEXTUAL TARGETS REMAIN PROVISIONAL RATHER THAN INDEPENDENTLY ADJUDICATED PRIVACY TRUTH.

RQ	Evidence	What it establishes
1	1,560 contextual cases; targeted ablation; 498 final presence rows	Agreement and detector contribution on declared targets.
1	33 installed receiver probes	Request disposition on tested paths.
2	Matched head/full continuation; 16-case guard	Paired gains/harms and conditional acceptance.
3	Historical cost; installed V3 activation; telemetry audit, simulation and operation	Bounded cost, delivery/accounting and content-privacy/estimation limits.

calls request [4] and assert its successful exclusive execution. Product outcomes are not semantic-only results.

Contextual targets and measures. The authored workload comprises 1,560 cases: 810 sensitive and 750 clean, with 260 per audience class, 35 scenario families and 24 audience-expression groups. Sensitivity, audience, category and allowed-action targets were prepared provisionally with assistant support and lack independent human adjudication. The cases were sealed for an earlier semantic comparison, then opened and reused for the present product comparison and ablation. These are development results on authored material, even where a subsequent protocol was fixed prospectively.

A nonzero sensitivity label or nonempty category set defines a positive binary prediction. Label-joint agreement requires exact sensitivity, audience and category-set matches; the *label-and-action joint* measure reported here additionally requires an allowed action. Allowed-action agreement is also measured alone. Under-restriction means an action below the allowed set, and over-restriction means an action above it. Actual receiver arrival is a separate systems observation. These definitions keep classification agreement, policy agreement and prevented transmission distinct.

Final annotation target. The once-only final endpoint contains 498 English rows from the Personally Identifiable Information Masking Benchmark (PIIMB) at revision `4a13e9f`: 236 annotation-positive and 262 annotation-negative [19]. Source inputs, all six packages and scoring were frozen before opening; no tuning, reselection or promotion followed. The same nonzero-sensitivity-or-category projection is evaluated through sole-semantic execution. Source annotation presence differs from contextual authorization and from PIIMB's official character-masking/F2 task. Its 464 observed source-group keys describe declared relationships, not proven independent templates.

Contextual uncertainty uses 2,000 paired whole-cluster draws, seed 42 and linear 95% percentile ranges, separately by family and audience expression. Ablation expression groups were reconciled post-collection. Final ranges use 2,000 paired source-group draws with the same seed/percentiles and row weighting. These unadjusted dependence sensitivities establish neither crossed independence, template separation nor population coverage; repeated calls and seeds add no independent situations.

B. Comparison and Retention Protocols

Historical and updated efficient/balanced/quality packages saw the same contextual cases through the product handler, with rules and policy fixed. Updated packages descend from frozen-encoder head fitting, joint encoder/pooler/head continuation, and audience-supervised rehearsal. Checkpoints were selected on validation alone before the sealed comparison. Historical input windows were 64/96/128 tokens with category threshold 0.38, versus 512 WordPieces and threshold 0.5 for updated packages. Their parameter counts also differ: 0.019/0.037/0.054 versus 4.785/9.596/17.494 million. These differences preclude isolating pretraining or architecture effects; repeatability passes add no cases.

The ablation fixes the updated balanced package, inputs and policy across full [2,3,4], no-rules [3,4] and no-NER [2,4], verifying execution and legitimate skips. It tests detector contributions here, not safe removal elsewhere.

RQ2 instead fixes the initial model and training exposure. Head-only and full-model treatments each started from byte-identical official pretrained balanced encoder/pooler weights and zero task heads, rather than either package in RQ1. Fits used Python 3.11, Torch 2.12/CUDA 13.0 and Transformers 4.57.6 on an RTX 5080. Seeds 42/43/44 received identical ordered 4,096 training exposures over 64 steps. Both minimized sensitivity and audience cross-entropy plus mean category binary cross-entropy, with dropout disabled. AdamW used head/encoder rates $10^{-3}/2 \times 10^{-5}$, zero weight decay, norm-one clipping and microbatch 4/effective batch 64. Full-model training enabled encoder optimization and consequently changed the joint-gradient clipping scope as well.

Complete training/guard group and normalized-text exclusions removed 480 validation rows, leaving 1,080 reused provisional cases across 25 families: 570 sensitive and 510 clean. Every terminal candidate and the unchanged base underwent native in-process sole-semantic assessment; assessment performance selected no checkpoint. The 16 separate guard cases required improved joint correctness and zero action harms against a single derived target action per case. The broader assessment instead accepts its declared action sets. Passing the guard and improving assessment agreement therefore answer different questions. No candidate was published: effective-artifact mappings describe what the rule would retain, conditional on a small guard and a weak zero-head control.

C. Enforcement, Cost and Telemetry Protocols

Thirty-three installed Linux Chromium probes used the built extension, standard native installer, supervisor, bridge and historical balanced detector. A local Hypertext Transfer Protocol Secure (HTTPS) receiver verified original bodies and late sends across forwarding, blocking, faults, cancellation and recovery. Outside resource windows, product/sole-semantic checks asserted actual execution. These are synthetic receiver tests, not authenticated-provider or cross-platform validation.

Cost measurement used three fixed synthetic cases, one per action, over fetch and XHR. Two fresh profiles per transport/action cell gave twelve profiles, each with a first decision, five warmups and thirty measured rules/NER requests: 360 measured decisions. Page latency spans the analysis request to its result, excluding the receiver response. Nearest-rank percentiles pool only two profile clusters per cell. Resource windows span browser launch through request/readiness checks, before teardown. Whole-container central processing unit (CPU) time and sampled charged memory include resident infrastructure, cache, driver, receiver and observer; charged memory is not process resident set size (RSS) or proportional set size (PSS) [20]. Sampling requested 100-ms intervals and recorded actual gaps. This historical V1 experiment contains no semantic processing, unprotected baseline or per-role attribution.

The V3 study crossed fetch/XHR with two fresh Linux Chromium profiles each, using "Please explain the steps in this process. Each retained its normal launcher, supervisor, bridge and model service across detectors-off, rules/NER, semantic and full arms; replicate two reversed order. Five warmups and thirty measurements per arm gave 80 warmups/480 timed requests. Page request-to-receiver completion permits within-profile median subtraction because all timed requests forwarded. The off baseline retains interception/infrastructure. Exact layers/packages were asserted on Torch 2.11 CPU, four threads, Python 3.12.3 and Chromium 153.0.8010.12, with telemetry disabled. Ten untimed post-cost probes covered three V3 tiers, older V1 balanced and return to default on both transports.

The telemetry audit enumerated exact rational kernels for a fingerprinted Windows CPython 3.11.9 build and enclosed logarithms with outward 100-digit arithmetic, conditional on independent uniform operating-system (OS) bits. Budgets 0.5, 1 and 2 cover the three declared settings. Estimator simulation crossed those budgets with $n = 8, 80, 800, 8000$, all five field domains and balanced-integer, point-mass and minimum-one 1%-target truth histograms: 180 prespecified cells, each repeated 10,000 times with seed 20261014. NumPy 1.26.4/PCG64 samples binary64 approximations of finite probabilities; the collector's nominal inverse and independent $[0, n]$ clipping produce the estimates. Large n pools installation-days, each still capped at eight default reservations. The estimand is the fixed contributor histogram, excluding population-sampling variation. Ledger checks separately exercise persistence, concurrent admission and failure accounting. No deployment-usefulness threshold was specified. A separate controlled Windows host study exercised the actual reporter/randomizer/ledger and a separate collector with SQLite persistence, using generated analysis-shaped fixtures without a detector. Twenty-three ledger fixtures tested caps, reopen, ordered inputs, outage/recovery and malformed packets. Synchronous report-call duration excludes background delivery; it is not combined with Linux browser timings. No real-user traffic was studied.

Artifact availability. Tracked recipes (`evaluation/`) and dated aggregate/provenance records (`paper/research/`) identify evaluated configurations through source/model/data fingerprints and environment receipts. They distinguish the product/ablation closure, including historical uncommitted dependencies, from matched continuation, final semantics, the pinned historical and standard-V3 client captures and telemetry. Some pinned assets, including run-specific checkpoints and raw run artifacts, remain local and ignored. A Git checkout therefore does not suffice for exact reruns; public distribution of those assets is not established.

TABLE II
FULL PRODUCT ON THE SAME PROVISIONAL CONTEXTUAL CASES. ENTRIES ARE HISTORICAL/UPDATED COUNTS. LABELS+ACTION DENOTES LABEL-AND-ACTION JOINT AGREEMENT; ACTION DENOTES ALLOWED-ACTION AGREEMENT (BOTH OUT OF 1,560). SENSITIVE AND CLEAN ARE CORRECTLY CLASSIFIED COUNTS OUT OF 810 AND 750, RESPECTIVELY. EVERY ARM HAD ZERO ERRORS.

Package	Labels+action	Action	Sensitive	Clean
Efficient	266/693	1029/1415	544/808	494/629
Balanced	160/717	1026/1409	771/802	318/631
Quality	226/879	1073/1421	598/810	468/641

V. RQ1: DETECTION AND ENFORCEMENT

A. Contextual Policy Agreement

The updated product packages more often selected the authored labels and an allowed action together. Label-and-action joint agreement increased by 27.37–41.86 percentage points, reaching 693–879 of 1,560 cases (Table II). All six runs completed without failed calls or layer errors. Efficient, balanced and quality gained 484/573/671 previously incorrect joint outcomes while losing 57/16/18 previously correct ones. Allowed-action gains were 389/389/348, with losses of 3/6/0. Quality therefore preserved every previously allowed-action match while losing eighteen label-and-action joint matches: preserving an intervention does not necessarily preserve its classification. These are improvements on reused provisional targets, and no package was promoted.

The execution records locate one important source of unnecessary restriction in the cascade. All updated packages over-restricted the same 139 cases: rules blocked 109 clean controls before later detectors could run and warned on thirty low-risk controls. Rules ran on every case and early-blocked 289 in total, leaving NER and semantics to execute on 1,271. A better downstream semantic model alone cannot resolve these shared failures. NER contributed zero converted PriVoke findings after supported-label filtering. That observation concerns the findings admitted into the policy; it does not imply that the loaded recognizer was inactive or detected no raw entities.

The fixed-package ablation tests this explanation (Table III), placing agreement beside under- and over-restriction. Removing rules corrected all 139 shared unnecessary restrictions but lost eighteen previously correct actions. The net gain of 121 actions therefore came with increased missed protection under the provisional policy. Paired action-rate gain ranges were 3.46–12.70 percentage points by family and 5.96–9.23 by audience expression. These separate descriptive ranges support the direction of the aggregate tradeoff without making the losses acceptable. No-NER outputs matched the full product,

TABLE III
BALANCED-PACKAGE ABLATION ON THE SAME 1,560 PROVISIONAL CASES. BRACKETS IDENTIFY REQUESTED LAYERS. ACTION AND JOINT COUNT ALLOWED-ACTION AND LABEL-AND-ACTION JOINT AGREEMENT; UNDER AND OVER COUNT ACTIONS BELOW AND ABOVE THE ALLOWED SET.

Configuration	Action	Joint	Under	Over
Full [2,3,4]	1409	717	12	139
No rules [3,4]	1530	972	30	0
No NER [2,4]	1409	717	12	139

TABLE IV
FINAL SEMANTIC ANNOTATION PRESENCE: 236 POSITIVE/262 NEGATIVE ROWS. HISTORICAL/UPDATED COUNTS; TRUE/FALSE POSITIVES (TP/FP), TRUE/FALSE NEGATIVES (TN/FN). ALL SIX ARMS HAVE ZERO ERRORS.

Package	TP	TN	FP	FN
Efficient	140/218	61/45	201/217	96/18
Balanced	141/221	65/35	197/227	95/15
Quality	167/190	51/57	211/205	69/46

consistent with zero converted findings on this fixture. Actual execution was verified; neither removal establishes that a detector is unnecessary on other workloads.

B. Agreement on a Separate Annotation Target

The final annotation-presence endpoint tested whether the same packages distinguished a different set of positive and negative targets. All 2,988 semantic-only calls completed without errors and verified sole execution of the requested layer. Updated recall reached 80.51–93.64%, while specificity remained only 13.36–21.76% (Table IV). Efficient and balanced lost specificity relative to their historical versions; quality's small increase remained uncertain. Frozen source-group resampling nevertheless gave paired net-accuracy gain ranges of 7.41–17.37, 5.35–15.03 and 1.02–10.71 percentage points, respectively. The apparently conflicting observations have a straightforward explanation: enough additional positives were recognized to improve average accuracy despite poor discrimination of annotation-negative rows.

The final result supplies a different check from the contextual product comparison. Here the target is source annotation presence and only the semantic layer runs; in the product comparison, provisional contextual labels and actions are assessed through a cascade. Their difference is not a controlled estimate of distribution shift or detector contribution. The low final specificity nevertheless establishes poor discrimination of annotation-negative rows under this declared projection. Neither endpoint supplies independently adjudicated truth about disclosure authorization.

C. Does a Decision Control Transmission?

The installed-browser capture tied decisions to transmission across thirty-three correctness probes: the receiver observed eighteen unchanged-body sends and fifteen zero-send outcomes. Correlated runtime/receiver records and late-arrival checks supported these outcomes without a decision broker.

TABLE V
MATCHED SEMANTIC-ONLY CONTINUATION ON 1,080 REUSED PROVISIONAL CASES. ENTRIES ARE HEAD-ONLY/FULL-MODEL COUNTS. LABELS+ACTION DENOTES LABEL-AND-ACTION JOINT AGREEMENT. THE UNCHANGED ZERO-HEAD BASE HAS ZERO SUCH MATCHES AND 480 ALLOWED-ACTION MATCHES. GUARD HARMS COMPARE EACH CANDIDATE WITH THAT BASE ON SIXTEEN SEPARATE CASES.

Seed	Labels+action	Allowed action	Guard harms
42	192/257	908/993	0/1
43	204/247	906/989	1/1
44	198/268	912/1003	0/1

Healthy full-pipeline and sole-semantic probes also verified actual execution, tying the observed request disposition to the tested detector path.

Fault probes showed why availability of a request path must be distinguished from preservation of its protection. Interrupting the parameter stream blocked semantic-only requests, whereas configured full-mode fallback ran rules and NER and forwarded the request. Restarting restored full execution; disabling protection or all layers forwarded requests by design. Fallback therefore maintained a reduced analysis path, but the capture does not establish equivalence to semantic protection. These completed observations remain separate from nine earlier failed captures, which were not pooled with them.

Answer to RQ1. PriVoke improved agreement with provisional contextual policy and enforced the declared request contract on the tested installed paths. The evidence supports those two bounded achievements, while action errors, original-content forwarding under WARN and weak specificity on the separate annotation-presence task limit the effectiveness claim. Independent contextual accuracy, provider compatibility, prevention on every sensitive prompt and practical usability remain unestablished.

VI. RQ2: UPDATES AND PREVIOUSLY PROTECTED BEHAVIOR

A. Improvement Subject to Retention

A candidate can improve measured agreement yet fail replacement criteria. RQ1 changes capacity, tokenizer, threshold and supervision together; matched RQ2 instead fixes base and exposure while enabling encoder optimization, separating learning gains from publication eligibility.

B. Matched Continuation and Guard Decisions

All six 64-step fits completed with finite parameters. HEADS preserved encoder tensors; FULL changed 39 with nonzero later gradients. All seven snapshots, including the base, completed 7,560 error-free native in-process semantic assessments. Each saw the same 1,080 situations: calls are not independent cases or deployed network measurements.

FULL improved joint and action agreement in every seed (Table V). For seeds 42/43/44, joint gains/losses were 79/14, 69/26 and 90/20; action gains/losses were 123/38, 124/41 and 110/19. Under-enforcement distance increased on 33/37/15 cases and over-enforcement on 17/13/17; these directional

TABLE VI
PAGE-OBSERVED RULES/NER DECISION LATENCY (MS). EACH CELL CONTAINS SIXTY MEASUREMENTS IN TWO FRESH PROFILES. MAX IS THE OBSERVED MAXIMUM. RECEIVER RESPONSE AND SEMANTIC PROCESSING ARE EXCLUDED.

Transport	Action	p50	p95	Max
fetch	ALLOW	35.0	62.5	73.6
fetch	WARN	31.9	36.1	59.4
fetch	BLOCK	27.9	31.6	33.2
XHR	ALLOW	30.8	35.8	57.7
XHR	WARN	31.3	34.6	59.2
XHR	BLOCK	28.5	32.2	34.1

counts are not a mutually exclusive harm union. Aggregate improvement therefore coexists with lost protection on these reused cases, without establishing a population frequency.

Against the unchanged base on sixteen guard cases, every FULL candidate introduced one action harm and failed. HEADS42/44 passed; HEADS43 failed. Failed candidates map to the base, forfeiting their assessment gains. No candidate in this experiment was published or promoted: the mappings are hypothetical. A small-guard pass cannot exclude other harms or overcome the weak zero-head baseline.

Answer to RQ2. Joint pretrained encoder/head continuation improved aggregate agreement over matched head-only training in all three seeds. Every FULL candidate nevertheless failed retention, as did head-only seed 43. This establishes a bounded learning gain and an enforceable acceptance decision, not a published improvement without sacrificed protection. Repeated narrow supervision is a plausible limitation; these experiments do not isolate overfitting as the cause of errors.

VII. RQ3: SYSTEMS AND PRIVACY OVERHEAD

A. Measured Systems Cost

The historical V1 measurements in Table VI end at a local rules/NER decision. All 240 ALLOW/WARN decisions ran both detectors; the 120 BLOCK decisions ended after rules and legitimately skipped NER. Differences between these cells therefore include different amounts of work. Medians range from 27.9 to 35.0 ms; maxima describe sixty requests in two profiles per cell. Receiver completion and semantic processing are excluded.

The 126.40-second resource windows accumulated 193.29 container CPU-seconds, including 0.69 observer CPU-seconds; CPU-seconds sum concurrent processor use. Maximum charged memory was 1,267.91 MiB, including services, cache, driver and receiver. Sampling gaps up to 622.84 ms can miss shorter spikes. Final-read/cleanup checks passed, but these totals do not isolate incremental overhead.

Table VII reports installed V3 balanced. Four within-profile median increments per arm span 26.1–26.7 ms for rules/NER, 9.4–17.0 ms for semantics and 31.9–33.5 ms for full detection; they are not per-request effects. All 480 timed attempts were ALLOW. With 80 warmups, 144 scaffold calls and ten post-cost checks, 714 calls produced 574 analysis RPCs and 712 unchanged-body arrivals, without execution errors. Quality BLOCKed the same benign prompt once per transport,

TABLE VII
INSTALLED V3 BALANCED REQUEST-TO-RECEIVER LATENCY (MS): SIXTY TIMED REQUESTS IN TWO PROFILES PER CELL. OFF RETAINS EXTENSION/NATIVE INFRASTRUCTURE; MAX IS OBSERVED, NOT A TAIL GUARANTEE.

Transport	Detectors	p50	p95	Max
fetch	Off	3.2	4.0	4.3
fetch	Rules/NER	29.6	33.8	92.4
fetch	Semantic	19.7	158.8	283.6
fetch	Full	35.4	172.8	202.1
XHR	Off	3.3	5.0	7.4
XHR	Rules/NER	29.9	31.8	100.5
XHR	Semantic	15.1	162.8	312.7
XHR	Full	35.9	175.8	341.0

TABLE VIII
ILLUSTRATIVE PRESPECIFIED CATEGORY CELLS ($K = 11$, EVENT $\varepsilon = 1$), EACH WITH 10,000 REPETITIONS. TRUTH, CLIPPED BIAS AND RMSE CONCERN THE FIRST CATEGORY IN COUNT UNITS. CLIP IS ITS CLIPPING FREQUENCY; SUM IS THE FREQUENCY THAT INDEPENDENTLY CLIPPED MARGINALS DO NOT SUM TO n (TOLERANCE 10^{-9}).

n	Truth	Bias	RMSE	Clip (%)	Sum (%)
80	7	+22.84	42.32	85.62	99.99
8000	727	+237.50	1028.47	28.32	99.78

producing both non-arrivals. Exact loading coexisted with an inappropriate decision; one repeated prompt cannot estimate a false-positive rate.

This isolates activation on one short input, excluding total extension-absent overhead and cold download/loading. Two profile clusters and maxima up to 341.0 ms support neither a service-level objective nor a deployment-tail guarantee. V1/V3 also differ in product code, dependencies and image, so cross-study speed differences cannot be attributed to the model alone.

B. Privacy Cost and Telemetry Utility

Auditing the finite implementation changed the privacy statement that could be made from the ideal formula alone. On the fingerprinted host, maximum event privacy loss exceeded nominal budgets 0.5, 1 and 2 by approximately $(1.294, 1.503, 1.226) \times 10^{-15}$. The excess is numerically tiny, but an exact nominal guarantee does not hold for that kernel. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same independent uniform-entropy assumptions. This bounds admitted event content under the audited conditions, without certifying other builds or deployment entropy.

Table VIII shows two prespecified balanced-integer category cells at event budget one and $n = 80/8000$. They illustrate clipping bias and root-mean-square error (RMSE), not utility acceptance or other cells. Absolute error increases with contributor count even as relative error can fall, and independently estimated marginals almost never sum to the total. More reports alone therefore do not establish reliable counts; deployment accuracy and acceptable application error remain unmeasured.

Accounting checks admitted eight default reservations and denied the ninth, retained the cap after process reopen and reset it on the next UTC day. Clock rollback and corrupt state failed closed; sixteen of forty concurrent epsilon-0.5 requests were admitted. Construction, queue and delivery failures still consumed their reservations. These outcomes support bounded admission under the tested state and clock conditions, but admission also selects which events appear. A constructed 32-event sequence contains 25% BLOCK overall; reversing its order changes BLOCK representation among the first eight contributors from 100% to 0%. Content randomization cannot repair this selection effect or conceal participation. A content-privacy bound consequently does not establish representative monitoring.

The controlled run offered 308 opportunities across 23 ledgers: 184 reservations, 168 accepted reports, sixteen submission failures and 124 pre-queue suppressions. Reopen, recovery and invalid-packet checks passed; noisy/estimated summaries survived collector restarts. Ordered ALLOW-then-BLOCK inputs exhausted budgets before BLOCK, reproducing selection. These fixtures are not users. Synchronous report-call p50/p95/max was 56.157/143.356/189.324 ms over all 308 opportunities, including suppressions. Reporting precedes response return but follows construction of the runtime elapsed field, which excludes that cost. Separate operator Health probes observed outage/recovery, not autonomous monitoring.

Answer to RQ3. Installed measurements now quantify resident V3 detector activation alongside historical infrastructure costs; total and cold-start overhead remain unresolved. Controlled telemetry verifies delivery/accounting, while finite privacy excess, estimation error and admission selection remain. The channel lacks performance fields and useful deployment monitoring is unmeasured; content privacy does not protect participation or network metadata.

VIII. Discussion and Limitations

A. What the System Evidence Supports

The cascade results make the placement of a detector part of the policy decision, not merely an implementation detail. An early rule BLOCK prevents semantic contribution, so improving a later model cannot repair that restriction. Removing rules corrected 139 unnecessary restrictions but lost eighteen previously correct actions. The implication is to review early rules against both restriction and protection costs, rather than adopt removal on its higher aggregate agreement alone. This conclusion is bounded by the provisional policy and tested fixture. Likewise, a policy-correct WARN releases original content: correct classification, appropriate intervention and prevented transmission require separate evidence.

Local sanitization exposes who controls continuation. Casper's acknowledged warnings permit a user decision; HaS and ProSan transform content to support continued use [8], [10], [11]. PriVoke's WARN instead proceeds without acknowledgement, while BLOCK leaves task completion unresolved. PriVoke neither masks prompts nor restores responses. Receiver compliance establishes this contract, not its desirability: authorized task completion, warning comprehension and reformulation effort remain unmeasured.

B. Ground Truth and Generalization Remain Open

Contextual integrity concerns appropriate information flows [6]; entity recognition supplies only part of that judgment. PriVoke's contextual targets are assistant-provisional, making independent contextual ground truth the largest scientific gap. Freezing the separate annotation-presence endpoint before opening limits post-selection on that endpoint, but its different target and sole-semantic scope do not validate contextual authorization. The authored-target gains and poor final specificity therefore answer different questions. Neither establishes human agreement, appropriate intervention or task utility.

Group/text exclusions reduce declared overlap, while paired cluster sensitivities address known dependence; neither establishes template independence, paraphrase exclusion or absence of pretraining exposure. The final endpoint contains 308 Nemotron, 89 AI4privacy, eighty Gretel, twenty Privy and one MAPA row. English synthetic/machine-generated cases do not represent other languages or organizational policies. Unadjusted intervals and three seeds give no population guarantee; confirmation after opening this endpoint requires a new protected successor.

C. Retention, Availability and Practical Utility

Matching the base and exposures limits RQ2 configuration confounding without retrospectively isolating the RQ1 package effect. Consistent with negative-flip research [7], every matched full-model candidate improved aggregate assessment yet failed the guard. Under the configured rule, casewise acceptance remains a veto that higher scores cannot cancel. No candidate in the matched study was published. The weak zero-head base and small guard limit what passing establishes, motivating independently reviewed retention cases and explicit harm criteria.

Blocking unavailable analysis can prevent disclosure while interrupting an authorized task; WARN instead proceeds without informed acknowledgement. Heuristic confidence is not calibrated joint correctness. Installed Linux evidence now covers historical V1 enforcement and standard V3 loading/cost, but the quality-tier benign-prompt BLOCKs show why installation is not accuracy validation. Authenticated providers, Windows, workers, attachments and other transports remain unverified. Authorized task completion, unnecessary-intervention cost, warning comprehension and reformulation effort were unmeasured, preventing a claim of practical utility or human benefit.

D. Measurement Boundaries and Next Steps

Matched resident arms now measure semantic/full detector activation, but instrumentation remains and no extension-absent or cold-load comparison establishes total overhead. RAPPOR motivates repeated-report accounting [13]; PriVoke's ledger requires retained state and a trusted clock, and its finite bound assumes independent uniform OS bits. Accounting does not remove estimator error or admission-order effects. Privacy, estimation and representative admission need separate

justification before operational use; the current deployment has too few concurrent users for a live-usage usefulness study. More users alone would not repair the missing fields, and no deployment-usefulness threshold is established. Lifetime categorical marginals and an operator Health probe cannot substitute for missing latency/error and usage-denominator instrumentation. Participation, delivery selection and metadata remain outside the bound; aggregates supply no corrected training labels.

The findings reveal barriers at each stage: early restrictions persist despite later model changes, improved candidates can remain ineligible, and admitted reports can remain unrepresentative. This motivates separate acceptance evidence, not a validated remedy. Contextual adjudication, broader receiver/retention checks, total-cost and user/task evidence remain necessary.

IX. Conclusion

PRIVOKE connects local prompt inspection, explicit request enforcement, version-bound guarded revision and separate randomized reporting in an implemented lifecycle. Its evaluation shows why those mechanisms need distinct acceptance evidence. Authored contextual agreement improved, yet action errors and poor specificity on the separate annotation-presence target remained. Installed-browser probes verified supported request handling, including original-content forwarding under WARN. Every matched full-model candidate improved aggregate agreement but failed retention, making average gain an insufficient basis for publication. Installed V3 measurements quantified resident activation cost while retaining an adverse quality-tier decision. Controlled telemetry verified delivery/accounting, without establishing live monitoring usefulness. The contribution is an inspectable mechanism and evidence of its limits; independent contextual accuracy, useful adaptation without lost protection and human benefit remain open.

AI Assistance

Artificial intelligence (AI) assistance was used to produce provisional contextual labels and support manuscript drafting. GPT-6 Sol assisted implementation and controlled testing, GPT-6 Luna assisted review, and GPT-6 Astra assisted manuscript review and rewriting. The provisional labels have not been established as independently adjudicated ground truth. Responsibility for the manuscript and its claims remains with the authors.

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